

1.3
3

a.)

p	q	$p \vee q$	$q \vee p$
T	T	T	T
T	F	T	T
F	T	T	T
F	F	F	F

b.)

p	q	$p \wedge q$	$q \wedge p$
T	T	T	T
T	F	F	F
F	T	F	F
F	F	F	F

1.3
7d

Ibrahim is not smart or not hardworking.

1.3
9b

$(p \rightarrow q) \rightarrow r$

$(\neg p \vee q) \rightarrow r$

$\neg(\neg p \vee q) \vee r$

$(p \wedge \neg q) \vee r$

1.3
12d

p	q	r	$p \vee q$	$p \rightarrow r$	$q \rightarrow r$	$(p \vee q) \wedge (p \rightarrow r) \wedge (q \rightarrow r)$	$[...] \rightarrow r$
T	T	T	T	T	T	T	T
T	T	F	T	F	F	F	T
T	F	T	T	T	T	T	T
T	F	F	T	F	T	F	T
F	T	T	T	T	T	T	T
F	T	F	T	T	F	F	T
F	F	T	F	T	T	F	T
F	F	F	F	T	T	F	T

1.3
15b

$p \rightarrow (p \vee q)$

$\neg p \vee (p \vee q)$ conditional-disjunction equivalence

$(\neg p \vee p) \vee q$ associative law

$T \vee q$ negation law

T domination law

1.3
15e

$$\neg(p \rightarrow q) \rightarrow p$$

$$\neg\neg(p \rightarrow q) \vee p$$

$$(p \rightarrow q) \vee p$$

$$(\neg p \vee q) \vee p$$

$$p \vee (\neg p \vee q)$$

$$(p \vee \neg p) \vee q$$

$$T \vee q$$

$$T$$

conditional-disjunction equivalence

double negation law

conditional-disjunction equivalence

commutative law

associative law

negation law

domination law

1.3
17

a)

P	q	$p \wedge q$	$p \vee (p \wedge q)$
T	T	T	T
T	F	F	T
F	T	F	F
F	F	F	F

b)

P	q	$p \vee q$	$p \wedge (p \vee q)$
T	T	T	T
T	F	T	T
F	T	T	F
F	F	F	F

1.3
19

$$(\neg q \wedge (p \rightarrow q)) \rightarrow \neg p$$

For this to be false, we must have $T \rightarrow F \equiv F$.

Thus, $\neg p \equiv F$, so $p \equiv T$.

We must have $\neg q \wedge (p \rightarrow q) \equiv T$, so $\neg q \equiv T$, meaning $q \equiv F$.

However, even though $p \equiv T$, $q \equiv F$ are the only possible truth values that could make the statement false, it's still true:

$$(\neg F \wedge (T \rightarrow F)) \rightarrow \neg T$$

$$(T \wedge (F)) \rightarrow F$$

$$F \rightarrow F$$

$$T$$

Thus the statement is a tautology.

$$\textcircled{1.3} \textcircled{65a} \quad \underbrace{(p \vee \neg q)}_T \wedge \underbrace{(\neg p \vee q)}_T \wedge \underbrace{(\neg p \vee \neg q)}_T$$

For this proposition to be true, each part of the "and" statement must be true. However, each of those is an "or" statement, so only one part of those must be true.

Consider $p \equiv F, q \equiv F: (F \vee \neg F) \wedge (\neg F \vee F) \wedge (\neg F \vee \neg F)$

Thus, this statement is satisfiable.

$$(F \vee T) \wedge (T \vee F) \wedge (T \vee T)$$

$$T \wedge T \wedge T$$

T

$\textcircled{1.3} \textcircled{66b}$ We can apply the same idea as before: we only need one proposition in each parentheses to be true for the entire statement to be true. Consider: $p \equiv F, s \equiv F, q \equiv T, r \equiv T$

$$(\neg p \vee \neg q \vee r) \equiv (\neg F \vee \neg T \vee T) \equiv T$$

$$(\neg p \vee q \vee \neg s) \equiv (\neg F \vee T \vee \neg F) \equiv T$$

$$(p \vee \neg q \vee \neg s) \equiv (F \vee \neg T \vee \neg F) \equiv T$$

$$(\neg p \vee \neg r \vee \neg s) \equiv (\neg F \vee \neg T \vee \neg F) \equiv T$$

$$(p \vee q \vee \neg r) \equiv (F \vee T \vee \neg T) \equiv T$$

$$(p \vee \neg r \vee \neg s) \equiv (F \vee \neg T \vee \neg F) \equiv T$$

$\textcircled{1.4} \textcircled{1}$ a) $P(0) \equiv "0 \leq 4"$ True b) $P(4) \equiv "4 \leq 4"$ True c) $P(6) \equiv "6 \leq 4"$ False

$\textcircled{1.4} \textcircled{2}$ $P(x) \equiv$ "The word x contains the letter a."

a) $P(\text{orange}) \equiv \text{True}$ b) $P(\text{lemon}) \equiv \text{False}$

c) $P(\text{true}) \equiv \text{False}$ d) $P(\text{false}) \equiv \text{True}$

(1.4) $P(x) \equiv$ "x spends more than five hours every weekday in class."

(5b) Domain: all students

$\forall x P(x) \equiv$ "Every student spends more than five hours each weekday in class."

(1.4) $C(x) \equiv$ "x is a comedian." $F(x) \equiv$ "x is funny"

Domain: all people

a) $\forall x (C(x) \rightarrow F(x)) \equiv$ "Every comedian is funny"

b) $\forall x (C(x) \wedge F(x)) \equiv$ "Every person is a comedian and funny."

c) $\exists x (C(x) \rightarrow F(x)) \equiv$ "There exists a person such that if they are a comedian, then they are funny."

d) $\exists x (C(x) \wedge F(x)) \equiv$ "There exists a person who is a comedian and funny."

(1.4) $R(x) \equiv$ "x is a rabbit" $H(x) \equiv$ "x hops" Domain: all animals

(8b) $\forall x (R(x) \wedge H(x)) \equiv$ "Every animal is a rabbit and hops"

(1.4) $P(x) \equiv$ "x can speak Russian" $Q(x) \equiv$ "x knows the computer language C++"

Domain: all students at your school

a) $\exists x (P(x) \wedge Q(x))$ b) $\exists x (P(x) \wedge \neg Q(x))$

c) $\forall x (P(x) \vee Q(x))$ d) $\neg \exists x (P(x) \vee Q(x)) \equiv \forall x (\neg P(x) \wedge \neg Q(x))$

(1.4) Domain: all integers

a) $\forall n (n+1 > n) \equiv \text{True}$ b) $\exists n (2n = 3n) \equiv \text{True}$ (consider $n=0$)

c) $\exists n (n = -n) \equiv \text{True}$ (consider $n=0$) d) $\forall n (3n \leq 4n) \equiv \text{False}$ (consider $n=-1$)

(1.4) Domain: $\{0, 1, 2, 3, 4\}$

(17c) $\exists x \neg P(x) = \neg P(0) \vee \neg P(1) \vee \neg P(2) \vee \neg P(3) \vee \neg P(4)$

1.4
25b "Not everyone is perfect" Domain: all people
 $P(x) \equiv "x \text{ is perfect}"$

$$\boxed{\forall x (\neg P(x)) \equiv \neg \exists x (P(x))}$$

1.4
33a "Some old dogs can learn new tricks." Domain: old dogs
 $T(x) \equiv "x \text{ can learn new tricks}"$

$$\exists x (T(x))$$

negation: $\boxed{\neg \exists x (T(x)) \equiv \forall x (\neg T(x))}$

"All old dogs can't learn new tricks"

1.4
36c $\exists x (-4 \leq x \leq 1)$

Let's rewrite the inside: $\exists x [(-4 \leq x) \wedge (x \leq 1)]$

negation: $\neg \exists x [(-4 \leq x) \wedge (x \leq 1)]$

$$\forall x \neg [(-4 \leq x) \wedge (x \leq 1)]$$

$$\forall x [\neg (-4 \leq x) \vee \neg (x \leq 1)]$$

$$\boxed{\forall x [(-4 > x) \vee (x > 1)]}$$

1.4
37 Domain: all integers

a) $\forall x (x^2 \geq x)$ No counterexample, this is true for all integers.

b) $\forall x (x > 0 \vee x < 0)$ Consider $x = 0$: $0 \not> 0$ and $0 \not< 0$

c) $\forall x (x = 1)$ Consider $x = 2$: $2 \neq 1$

1.4
38b Domain: all real numbers

$\forall x (x^2 \neq 2)$ Consider $x = \sqrt{2}$: $x^2 = 2$ ($\sqrt{2}$ is a real number)

1.4
43a $F(x,y) \equiv$ "Disk x has more than y kilobytes of free space."
 $S(x) \equiv$ "Mail message x can be saved"

$$(\exists x [F(x,10)]) \rightarrow \exists z [S(z)]$$

1.4
53 Consider $P(x) \equiv$ " x is odd" $Q(x) \equiv$ " x is even"
Domain: all integers

$\exists x P(x) \wedge \exists Q(x) \equiv$ True, because there do exist odd and even integers

$\exists x (P(x) \wedge Q(x)) \equiv$ False, because no integer is both odd and even at the same time.

1.4
55 a) $\exists! x P(x) \rightarrow \exists x P(x) \equiv$ True

If you know a unique one exists, then you must know one exists.

b) $\forall x P(x) \rightarrow \exists! x P(x) \equiv$ False (usually). True under specific circumstances
True if the domain has only one element or if $\forall x P(x)$ is false.
Otherwise, it is false.

c) $\exists! x \neg P(x) \rightarrow \neg \forall x P(x) \equiv$ True

If there is exactly one x such that $P(x)$ is false then it can't be true that $P(x)$ is true for every x . Thus, $\neg \forall x P(x)$ must follow.

1.4
62c $P(x) \equiv$ " x is a clear explanation"

$Q(x) \equiv$ " x is satisfactory"

$R(x) \equiv$ " x is an excuse"

Domain: all English text

"Some excuses are not clear explanations."

$$\exists x (R(x) \wedge \neg P(x))$$

1.4
1.3c $P(x) \equiv "x \text{ is a baby}"$ $Q(x) \equiv "x \text{ is logical}"$
 $R(x) \equiv "x \text{ is able to manage a crocodile}"$ $S(x) \equiv "x \text{ is despised}"$
 "Illogical people are despised"

$$\forall x (\neg Q(x) \rightarrow S(x))$$

1.5
1 Domain: all real numbers

a) $\forall x \exists y (x < y) \equiv$ "For any real number x , there exists a real number y greater than x ." (aka there is no largest real number)

b) $\forall x \exists y (((x \geq 0) \wedge (y \geq 0)) \rightarrow (xy \geq 0)) \equiv$ "For any real number x , there exists a real number y such that if x is nonnegative and y is nonnegative, then xy is nonnegative."

(aka the product of nonnegative numbers is nonnegative.)

c) $\forall x \forall y \exists z (xy = z) \equiv$ "For any real numbers x and y , there exists a real number z such that $xy = z$ "

(aka the real numbers are closed under multiplication.)

1.5 $P(x, y) \equiv$ "Student x has taken class y ."

4a Domain of x : all students in your class

Domain of y : all computer science courses at your school

$\exists x \exists y P(x, y) \equiv$ "There exists a student in your class x and a computer science course at your school y such that x has taken y ."

(aka Some student has taken some computer science course"

1.5
6b $C(x, y) \equiv$ "student x is enrolled in class y "

Domain of x : all students at your school

Domain of y : all classes being given at your school.

$\exists x C(x, \text{Math 695}) \equiv$ "There exists a student at your school x enrolled in class Math 695" (aka Someone is enrolled in Math 695)

1.5
11 $S(x) \equiv$ "x is a student" $F(x) \equiv$ "x is a faculty member"

$A(x, y) \equiv$ "x has asked y a question"

Domain: all people associated with your school.

a) $A(\text{Lois, Professor Michaels})$

b) $\forall x [S(x) \rightarrow A(x, \text{Professor Gross})]$

c) $\forall x [F(x) \rightarrow (A(x, \text{Professor Miller}) \vee A(\text{Professor Miller}, x))]$

d) $\exists x \forall y [S(x) \wedge (F(y) \rightarrow \neg A(x, y))]$

e) $\exists y \forall x [F(y) \wedge (S(x) \rightarrow \neg A(x, y))]$

f) $\exists x \forall y [S(x) \wedge (F(y) \rightarrow A(x, y))]$

g) $\exists x \forall y (F(x) \wedge (F(y) \rightarrow A(x, y)))$

h) $\exists x \forall y [S(x) \wedge (F(y) \rightarrow \neg A(y, x))]$

1.5
21 Domain: all integers $\forall x \exists a \exists b \exists c \exists d [x = a^2 + b^2 + c^2 + d^2]$

1.5
23 Domain: all real numbers

a) $\forall x \forall y [(x < 0) \wedge (y < 0) \rightarrow (xy > 0)]$

b) $\forall x [x - x = 0]$

c) $\forall x \exists a \exists b [a \neq b \wedge \forall c (c^2 = x \leftrightarrow (c = a \vee c = b))]$

d) $\forall x [(x < 0) \rightarrow \neg \exists y (x = y^2)]$

1.5
24d The product of two numbers is nonzero if and only if both factors are nonzero. (2)